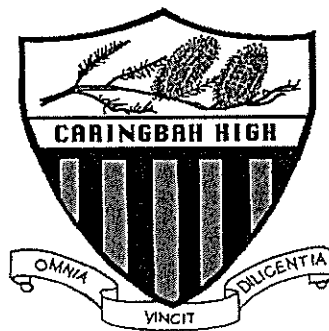


# *Caringbah High School*

## **Year 11 Mathematics Extension 1 Semester 2 Examination 2014**



**Time Allowed:  $1\frac{1}{2}$  hours + 5 minutes reading time**

### **Instructions**

- All questions must be attempted.
- Start each question on a new page.
- Answers without necessary working may not attract full marks.
- Marks may not be awarded for careless or badly arranged work.
- Diagrams and working which lead to a correct solution may attract some marks.
- Approved calculators may be use



**Question 1** (10 marks) Use the Question 1 Writing Booklet.

(a) Find  $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$  1

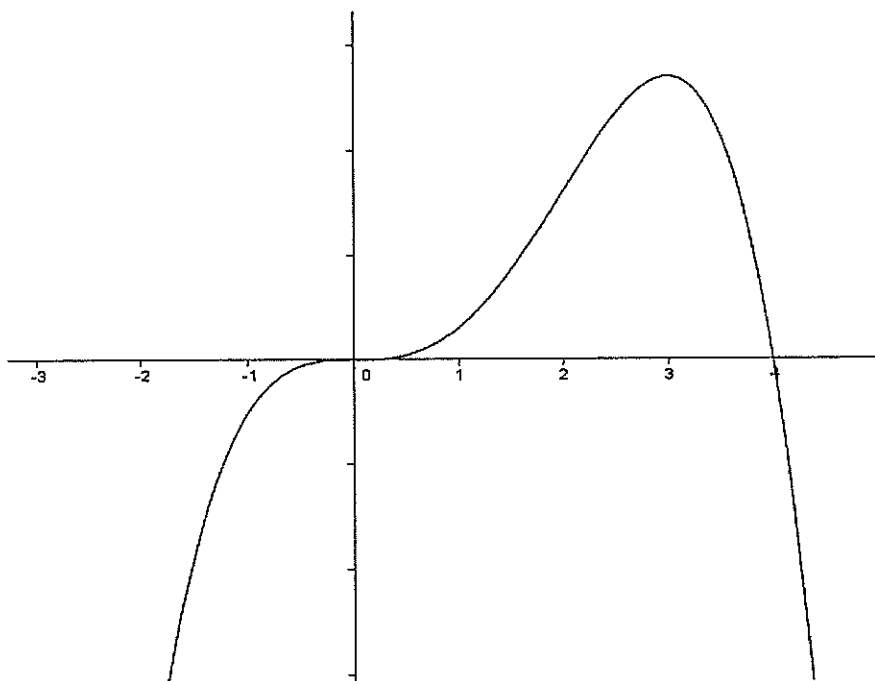
(b)

(i) Differentiate  $y = \frac{x^2 - 3}{2x - 3}$  2

(ii) Find the value of  $f'(2)$  1

(iii) Hence find the equation of the tangent to the function at this point. 2

(c) Draw the graph of the gradient function to the following curve. 2



(d) Determine whether the following quadratic equation has equal rational roots, unequal rational roots, unequal rational roots or no real roots. 2

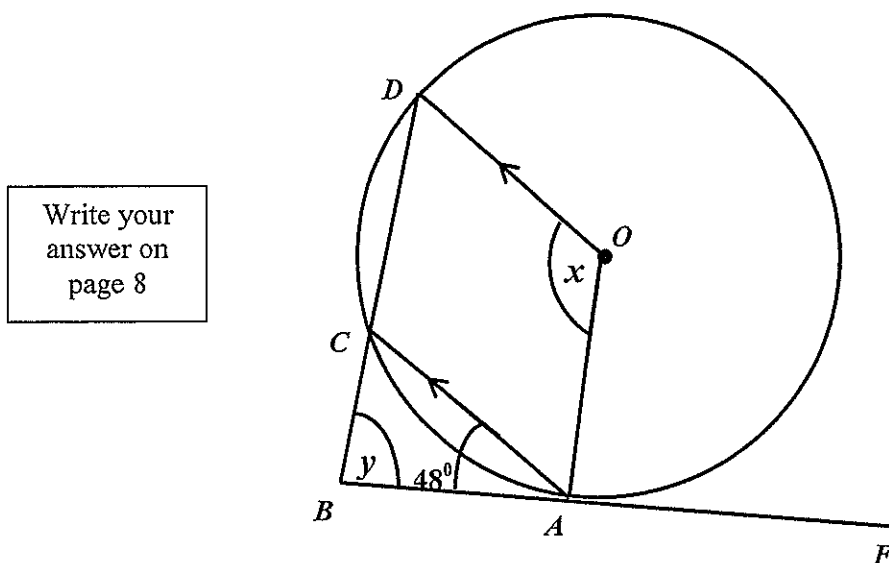
$$4x^2 + 4x + 1 = 0$$

**Question 2** (10 marks) Use the Question 2 Writing Booklet.

- (a) Find the values of  $a$  so that the line  $x + a = 0$  is a tangent to the circle centre  $(1, -2)$  and radius 3 units. 2
- (b) Find the point that divides the interval  $(-2, 3)$  and  $(4, -2)$  externally in the ratio 1:2. 2
- (c) Given three points  $A(-1, 1)$   $B(1, 2)$  and  $C(2, -3)$
- (i) Find the exact distances  $AC$  and  $AB$  2
- (ii) Find the angle between the lines  $AB$  and  $AC$ . 2
- (iii) Find the area of the triangle formed by joining  $ABC$ . 2

Question 3 (10 marks) Use the Question 3 Writing Booklet.

- (a)  $BF$  is a tangent to the circle centre  $O$  at  $A$ . 3  
 $\angle CAB = 48^\circ$ ,  $\angle DOA = x^\circ$ ,  $\angle ABC = y^\circ$   
 Find the values of  $x$  and  $y$  giving reasons.

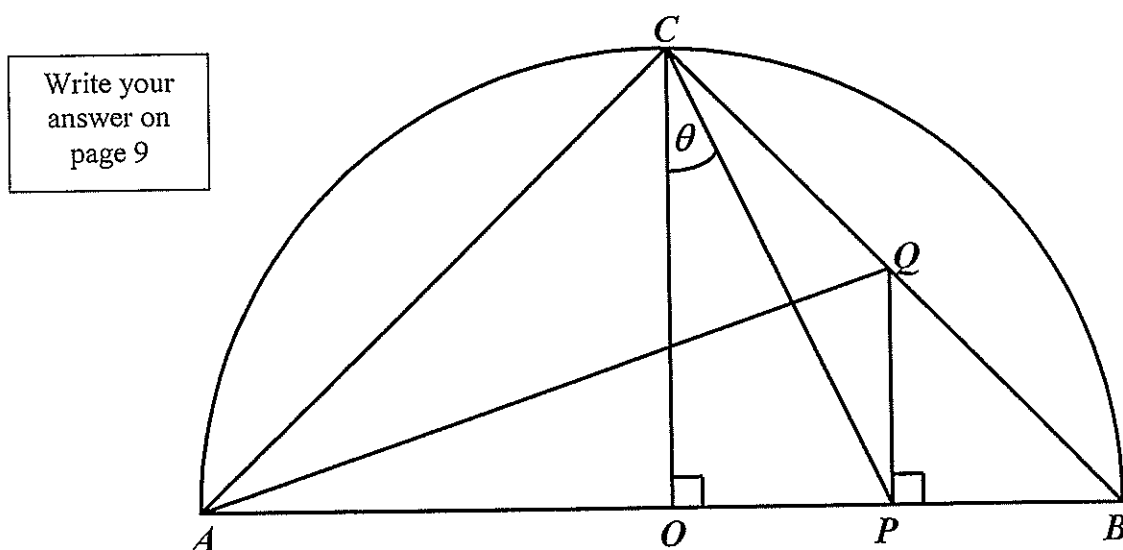


- (b) Solve  $\frac{3x}{3x-4} < 3$  2
- (c) The general form of the line through the point of intersection  $M$  of two lines  
 $2x - 2y + 5 = 0$  and  $x + 3y + 2 = 0$  is  
 $l: 2x - 2y + 5 + k(x + 3y + 2) = 0$
- (i) Show that the gradient of  $l$  is  $\frac{2+k}{2-3k}$  1
- (ii) Hence find the equation of the line through  $M$  parallel to the line  
 $3y - x - 5 = 0$ . Write your equation in general form. 2
- (d) Shade in the region for which both the following inequalities hold simultaneously. 2  
 $y \geq x^2$ ,  $y < \frac{8}{x}$

**Question 4** (10 marks) Use the Question 4 Writing Booklet.

(a) Differentiate by first principles  $y = \sqrt{x}$  3

(b) The point  $O$  is the centre of the semi-circle.  $OC$  and  $QP$  are perpendicular to  $AB$  and  $\angle OCP = \theta$ . 3  
 Prove that  $\angle PCA = \angle PQA = 45^\circ + \theta$



(c) Find the value of  $m$  if the equation  $x^2 - 3mx + (m + 3) = 0$

(i) has the sum of the roots equal to their product. 2

(ii) has one root the reciprocal of the other. 2

**Question 5** (10 marks) Use the Question 5 Writing Booklet.

(a) If  $f(x) = \begin{cases} x & x \geq 0 \\ x^3 & x < 0 \end{cases}$

(i) Sketch the curve for  $-2 \leq x \leq 2$  2

(ii) Find  $f(2) - f(-2)$  1

(iii) Is  $f(x)$  able to be differentiated at  $x = 0$ ? Give a reason as to why it is or is not. 1

(b)  $f(x) = \frac{x}{x^2 - 1}$

(i) Find the domain and range for  $f(x)$ . 2

(ii) Explain why  $f(x)$  is odd, even or neither. 1

(iii) Sketch the function showing any asymptotes and intercepts. 2

(c) Show that  $\lim_{x \rightarrow \infty} \frac{x^3 + 2x^2 - 3}{2x^3 + x}$  is  $\frac{1}{2}$ . 1

**Question 6** (10 marks) Use the Question 6 Writing Booklet.

(a) Differentiate and express your answers in the most factorised form.

(i)  $(x^2 - 3)^2(x^2 + 2)$  2

(ii)  $\left(x + \frac{1}{x}\right)^3$  2

(b) For the equation  $x^2 + rx + p = 0$ . If one root is twice the other show that  
 $2r^2 = 9p$  2

(c) Solve the equation  $x^4 + 6x^3 + 6x^2 - 9x - 4 = 0$  by expressing it in the form  
 $(x^2 + ax)^2 + b(x^2 + ax) + c$  and hence solve the equation  
 $x^4 + 6x^3 + 6x^2 - 9x - 4 = 0$  4

**End of Paper**



Student Number

**Examination:**.....

b)  $y = \frac{x^2 - 3}{x - 3}$

$$u = x^2 z$$

$$v = 2x - 3$$

$$\mu = 2\sigma$$

$$v = 2$$

$$y' = \frac{(2x-3)2x - 2(x^2-3)}{(2x-3)^2}$$

$$= \frac{4x^2 - 6x - 2x^2 + 6}{(2x-7)^2}$$

$$= \frac{2x^2 - 6x + 6}{(3(-3))^2}$$

ii)  $f'(z) = \frac{8-12z+6}{z^2} = 2$

111) ~~111~~  $f(2) = \frac{4-3}{4-3} = 1$

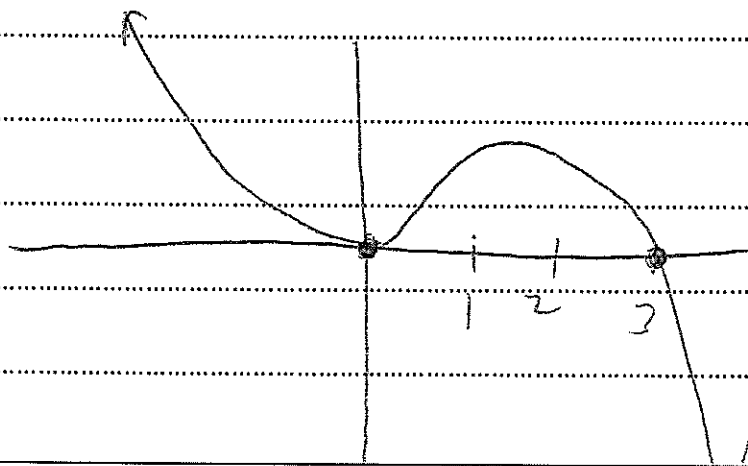
$$y - 1 = 2(x - 2)$$

$$y = 2x - 3$$

$$d) \quad 4x^2 + 4x + 1 = 0$$
$$(2x + 1)^2 = 0$$

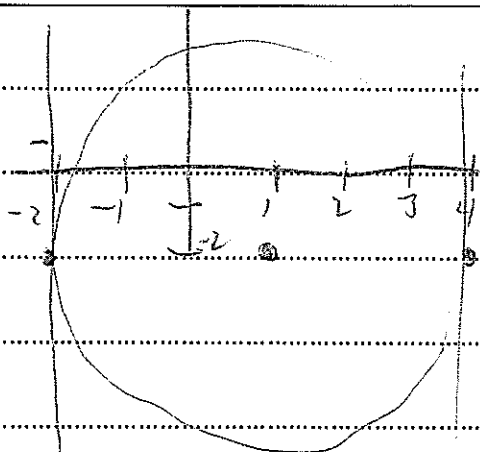
$$x = -\frac{1}{2}$$

So equal rational roots.



Q2

a)



$$x = 4 \text{ or } x = -2$$

80

$$x - 4 = 0$$

$$x + 2 = 0$$

$$a = -4, 2$$

b)

$$(-2, 3)$$

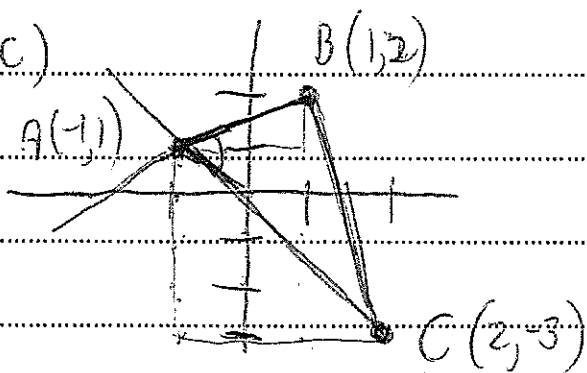
$$(4, -2)$$

$$-1 : 2$$

$$\left( \frac{-1 \times 4 + 2 \times -2}{1}, \frac{-1 \times -2 + 2 \times 3}{1} \right)$$

$$= (-8, 8)$$

c)



$$i) AB = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$ii) AC = \sqrt{3^2 + 4^2} = 5$$

$$ii) m AB = \frac{1}{2}$$

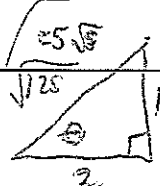
$$m AC = -\frac{4}{3}$$

$$\tan \theta = \left| \frac{\frac{1}{2} + \frac{4}{3}}{1 - \frac{2}{3}} \right| = \frac{\frac{11}{6}}{\frac{1}{3}} = 5\frac{1}{2}$$

$$\theta = 80^\circ (79^\circ 42')$$

$$iii) \text{Area} = \frac{1}{2} \times 5 \times \sqrt{5} \times \sin 79^\circ 42' = 5.5 u^2$$

$$\sin \theta = 5\sqrt{5}$$



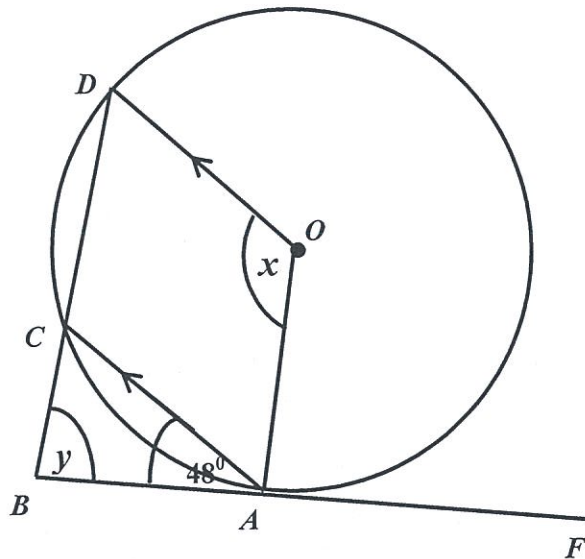
Answer Sheet for Q3 (a)

$BF$  is a tangent to the circle centre  $O$  at  $A$ .

3

$\angle CAB = 48^\circ$ ,  $\angle DOA = x^\circ$ ,  $\angle ABC = y^\circ$

Find the values of  $x$  and  $y$  giving reasons.



$$\angle CAO = 42^\circ \text{ (radius \& tangent = } 90^\circ \text{)}$$

$$\angle DOA = x = 138^\circ \text{ (co-interior } AC \parallel OD \text{)}$$

$$\text{reflex } \angle DOA = 222^\circ$$

$$\therefore \angle DCA = 111^\circ \text{ (}\angle \text{at centre} = 2 \times \angle \text{at circumf.)}$$

$$\therefore y = 63^\circ \text{ (external angle = sum of opposite interior } \angle \text{ s in } \triangle \text{)}$$

Q3

3

$$b) \frac{3x}{3x-4} < 3 \quad x \neq \frac{4}{3}$$

$$\frac{3x(3x-4)^2}{(3x-4)^2} < 3(3x-4)^2$$

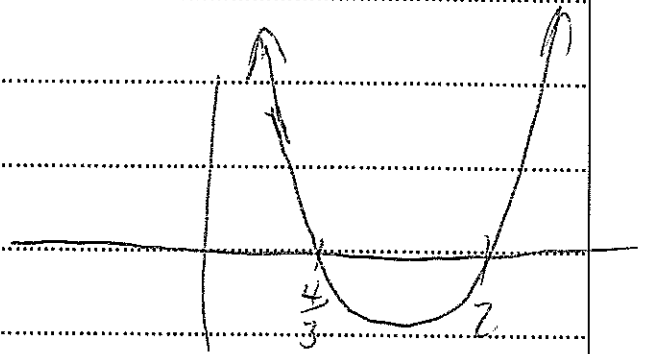
$$0 < 3(3x-4)^2 - 3x(3x-4)$$

$$0 < 3(3x-4)(3x-4-x)$$

$$0 < 3(3x-4)(2x-4)$$

$$0 < 6(3x-4)(x-2)$$

$$\therefore x < \frac{4}{3} \text{ and } x > 2$$



$$c) i) 2x - 2y + 5 + kx + 3ky + 2kz = 0$$

$$(k+2)x - (2-3k)y + 5 + 2kz = 0$$

$$\therefore y = \frac{k+2}{2-3k}x + \frac{5+2k}{2-3k}$$

$$ii) 3y = x + 5 \quad y = \frac{1}{3}x + \frac{5}{3}$$

$$m = \frac{1}{3}$$

$$\frac{k+2}{2-3k} = \frac{1}{3}$$

$$3k+6 = 2-3k$$

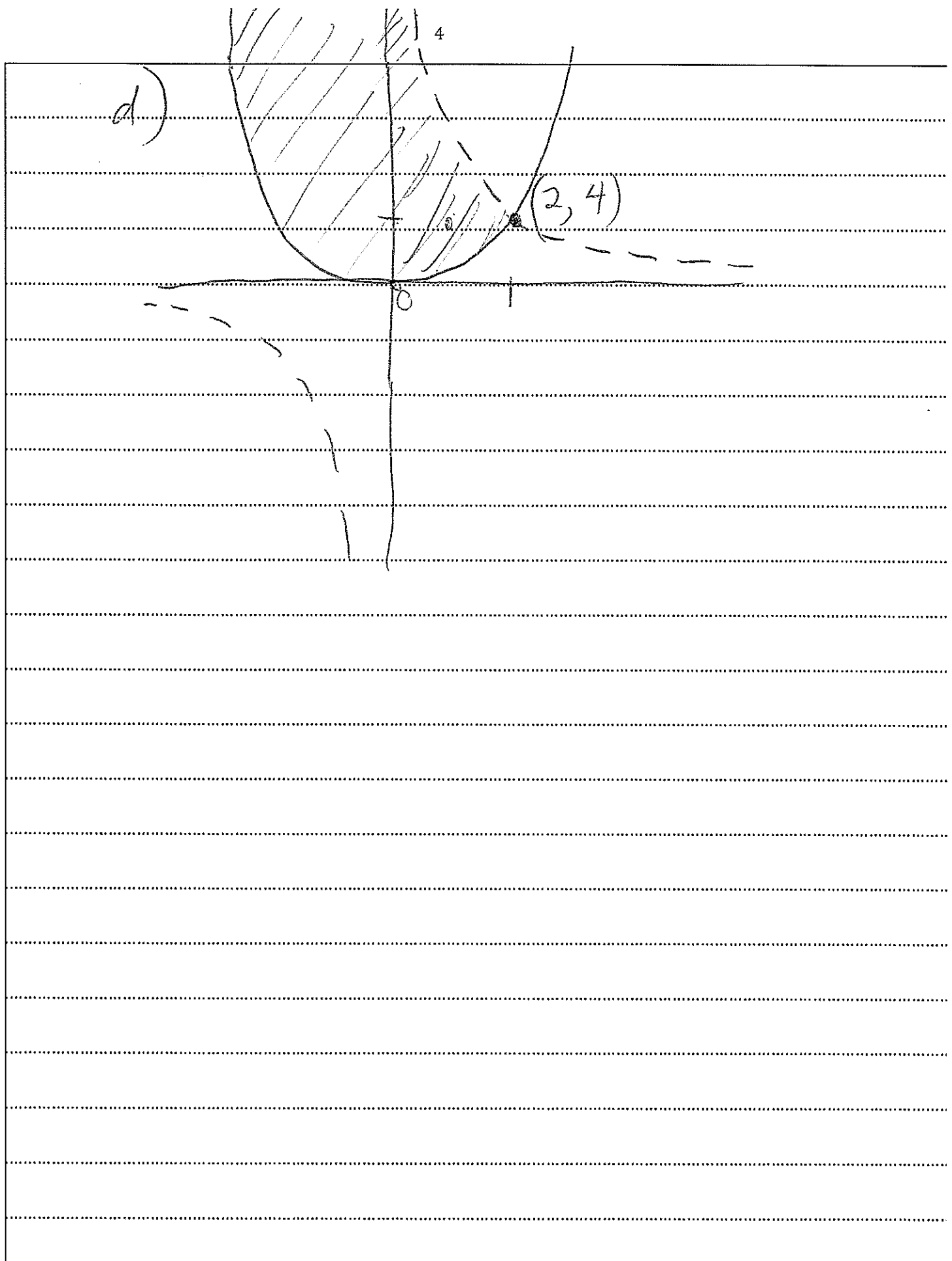
$$6k = -4$$

$$k = -\frac{2}{3}$$

$$\therefore 2x - 2y + 5 - \frac{2}{3}(x + 3y + 1) = 0$$

$$6x - 6y + 15 - 2x - 6y - 4 = 0$$

$$4x - 12y + 11 = 0$$



You may ask for an extra Writing Booklet if you need more space to answer this question.





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Student Number

Examination:.....

Q4

$$a) \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \times \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$$

$$= \lim_{h \rightarrow 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \frac{1}{2\sqrt{x}}$$

b) ~~see~~ ~~see~~

$$c) x^2 - 3xm + (m+3) = 0$$

$$i) \alpha + \beta = 3m \quad \alpha\beta = m+3$$

$$3m = m+3$$

$$2m = 3$$

$$m = \frac{3}{2}$$

$$ii) \alpha \times \frac{1}{\alpha} = m+3$$

$$1 = m+3$$

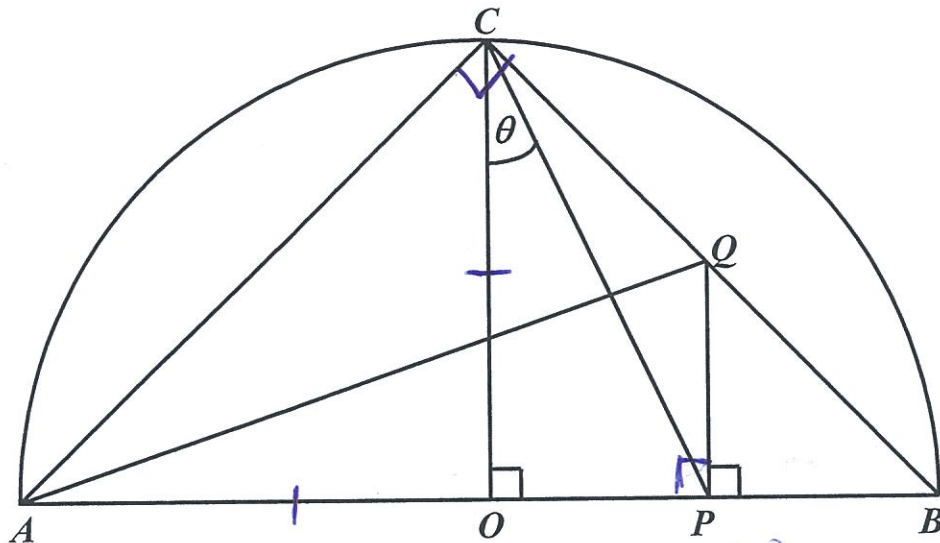
$$m = -2$$

Answer Sheet for Q4 (b)

The point O is the centre of the semi-circle. OC and QP are perpendicular to AB and  $\angle OCP = \theta$ .

3

Prove that  $\angle PCA = \angle PQA = 45^\circ + \theta$



$$\angle ACB = 90^\circ \text{ (}\Delta \text{ in a semicircle)}$$

ACQP is a cyclic quad as opposite  $\angle$ 's are supplementary

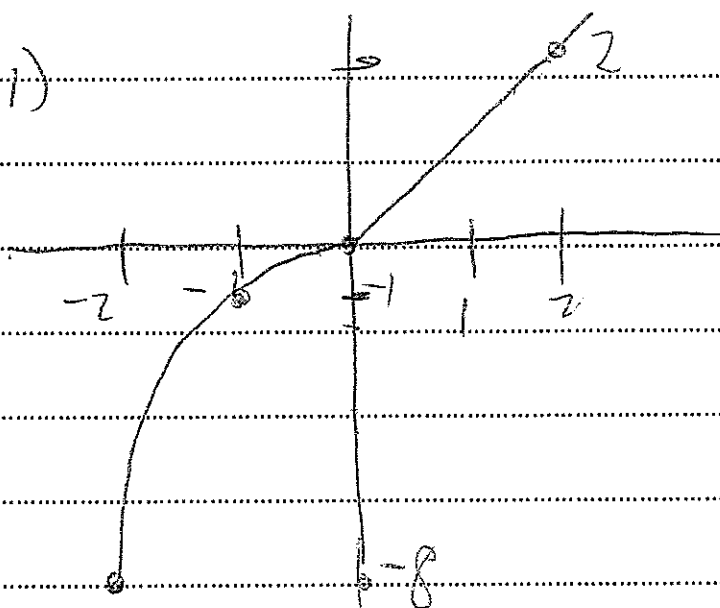
$$\therefore \angle ACP = \angle AQP \text{ (}\angle \text{'s in same segment)}$$

$\Delta AOC$  is isosceles +  $\angle$  at centre (AO, OC radii)

$$\therefore \angle ACO = 45^\circ$$

$$\therefore \angle PCA = 45^\circ + \theta$$

Q5. i)



$$ii) f(2) - f(-2) = 2 - (-8) = 10$$

$$iii) \text{ No as } f'(x) = 3x^2 \text{ for } x < 0 \text{ and } f'(x) = 1 \text{ for } x > 0$$

$$\lim_{x \rightarrow 0^-} = 0 \quad \lim_{x \rightarrow 0^+} = 1 \quad \text{So different values}$$

at  $x = 0$  It is a sharp corner so no tangent can be drawn that is unique.

$$b) f(x) = \frac{x}{x^2 - 1}$$

$$i) \text{ } D: x \neq \pm 1 \text{ all real } x \text{ except } x \neq \pm 1$$

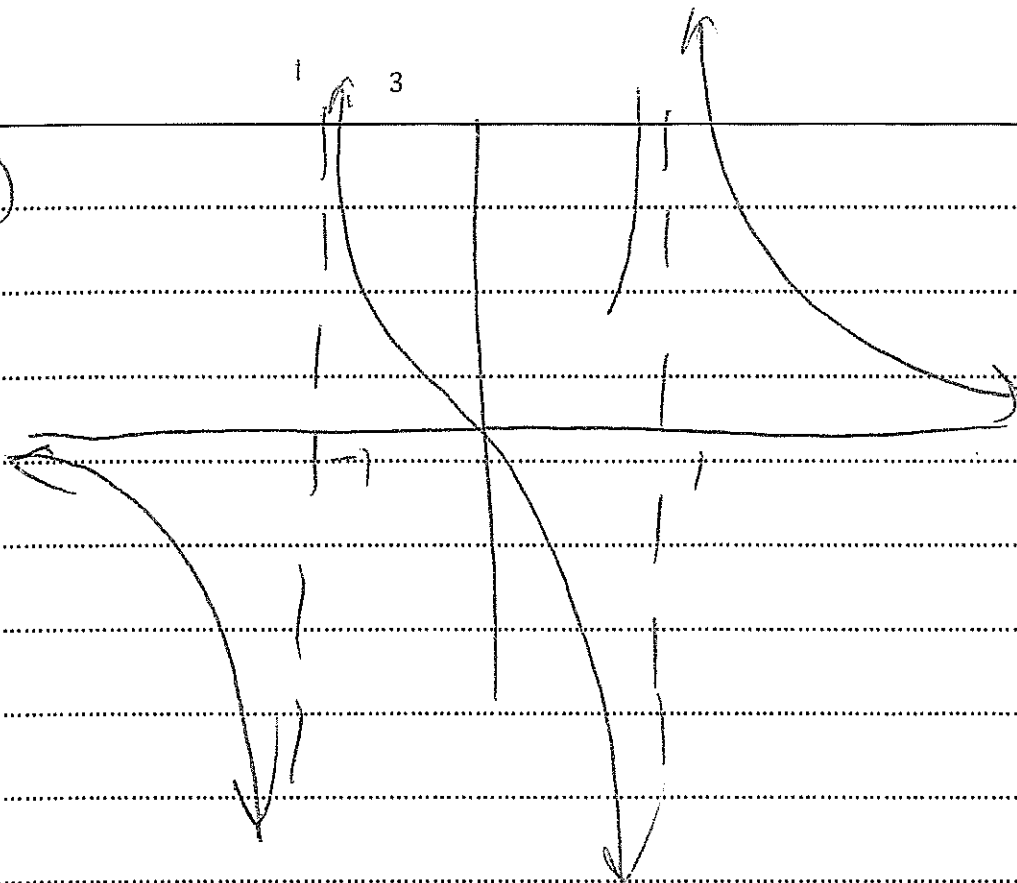
$$R: \text{ All real } y.$$

$$ii) f(-x) = \frac{-x}{(-x)^2 - 1} = \frac{-x}{x^2 - 1} = -f(x)$$

$\therefore f(x)$  is odd



iii)



$$c) \lim_{x \rightarrow \infty} \frac{\frac{x^3}{x^3} + \frac{x^2}{x^3} - \frac{3}{x^3}}{\frac{2x^3}{x^3} + \frac{x}{x^3}}$$

$$= \lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x} - \frac{3}{x^3}}{2 + \frac{1}{x^2}}$$

$$= \frac{1}{2}$$

Q6. a) i)  $(x^2-3)^2(x^2+2)$

$$\begin{aligned} y &= 2 \times 2x(x^2-3)(x^2+2) + 2x(x^2-3)^2 \\ &= 2x(x^2-3)(2x^2+4+x^2-3) \\ &= 2x(x^2-3)(3x^2+1) \end{aligned}$$

ii)  $3\left(x+\frac{1}{x}\right)^2\left(\frac{1}{x^2}\right)$

b)  $x^2+rx+p=0$

$$\alpha+\beta=-r \quad \alpha\beta=p$$

But  $\alpha=2\beta$

$$\therefore 3\beta=-r \quad 2\beta^2=p$$

$$\beta = -\frac{r}{3}$$

$$\text{So } 2\left(-\frac{r}{3}\right)^2 = p$$

$$2r^2 = 9p$$





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Student Number

Examination:.....

$$c) \quad x^4 + 6x^3 + 6x^2 - 9x - 4 = (x^2 + ax)^2 + b(x^2 + 0x) + c$$

$$x = 0 \quad c = -4$$

$$2a^3c^3 = 6x^3$$

$$\therefore 2a = 6$$

$$a = 3$$

$$x = 1$$

$$0 = (1+3)^2 + b(1+3) - 4$$

$$0 = 16 + 4b - 4$$

$$4b = -12$$

$$b = -3$$

So

$$(x^2 + 3x)^2 - 3(x^2 + 3x) - 4 = 0$$

$$u^2 - 3u - 4 = 0$$

$$(u-4)(u+1) = 0$$

$$x^2 + 3x - 4 = 0$$

$$x^2 + 3x + 1 = 0$$

$$(x+4)(x-1) = 0$$

$$x = \frac{-3 \pm \sqrt{5}}{2}$$

$$x = 1, -4$$

are the solutions

## Year 11 S2

## Mathematics Ext 1 Outcome Sheet

Name: \_\_\_\_\_

| Question | Geometry | Functions and Graphs | Quadratic Function | Calculus            | Coordinate | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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